Introduction

Data Redundancy

Redundancy is the duplication of data or the storing of the same data in more than one place.

The **problems**, other than redundancy, associated with the **no database approach** to processing data include:

- difficulties accessing related data
- limited security features to protect data from access by unauthorized users
- limited ability for multiple users to update the same data at the same time
- size limitations.

Database Background

Entity

An entity is a person, place, object, event, or idea for which you want to store and process data.

Attribute

An attribute, which is also called a **field** or **column** in many database systems, is a characteristic or property of an entity.

Example: Customer has name, street, city, etc.

Database Background

Data file

- File used to store data
- Computer counterpart to ordinary paper file

Database

A database is a structure that can store information about:

- Multiple types of entities
- Attributes of those entities
- Relationships between the entities

Relationships

Relationship

A relationship is an association between entities

One-to-many relationship

A one-to-many relationship exists between two entities when each row in the first entity matches many rows in the second entity and each row in the second entity matches only one row in the first entity.

Example: An advisor has many advisees but an advisee has only one

Entity-relationship (E-R) diagram

Entity-relationship (E-R) diagram

An E-R diagram is a **v**isual way to represent a database

In an E-R diagram

- Rectangles represent entities
- Lines represent relationships between connected entities

Database Management Systems

Database management system (DBMS)

A database management system is a program, or collection of programs, through which users interact with a database

Popular DBMSs: Access, Oracle, DB2, MySQL, and SQL Server

Database design

Database design is determining the structure of the required database

Database Management Systems

Forms

A form is a screen objects used to maintain, view, and print data from a database

- DBMS creates forms that Premiere Products needs

Reports

The DBMS creates reports based on user's answers about the desired content.

Advantages of Database Processing

1. Getting more information from the same amount of data
2. Sharing data
3. Balancing conflicting requirements
   – **Database administrator** or **database administration (DBA**): person or group in charge of the database
4. Controlling redundancy
5. Facilitating consistency

Advantages of Database Processing

6. Improving integrity
   - **Integrity constraint**: a rule that data must follow in the database
7. Expanding security
   - **Security**: prevention of unauthorized access
8. Increasing productivity
9. Providing data independence
   - **Data independence**: can change structure of a database without changing the programs that access the database

Disadvantages of Database Processing

1. Larger file size
2. Increased complexity
3. Greater impact of failure
4. More difficult recovery

Database Background

Data Independence

Data independence is the property that lets you change the structure of a database without requiring you to change the programs that access the database.

With data independence, you easily can change the structure of the database when the need arises.

Security

Security is the prevention of access to the database by unauthorized users.